Lecture 25

Topological band theory

Objectives. *Lecture 24 attached a line bundle to an isolated simple eigenvalue. For a periodic crystal, the parameters are momenta on a torus. The normalized curvature integral on that torus is an integer. For a two-band model we compute it as a degree, then use the cited Hall-response law to relate this integer to a measured conductance.*

By the end of this lecture you will be able to:

- Construct a Bloch eigenline bundle for a smooth isolated simple band.
- Prove integrality of its normalized curvature flux on the oriented momentum torus.
- Compute the lower-band Chern number and gap closings of the QWZ model.
- State the filled-band Hall formula with its physical hypotheses and component convention.

Unit 25A Bloch bundles and the QWZ degree

25.1 Bloch Hamiltonians and the momentum torus

Consider a square lattice with cell label $r \in \mathbb{Z}^2$ and N orbitals in each cell. Use lattice spacing one and a fixed orthonormal orbital basis. A one-particle wavefunction has a column $\psi_r \in \mathbb{C}^N$ at each cell. A translation-invariant finite-range Hamiltonian is given by

$$(H\psi)_r = \sum_{R\in\mathbb{Z}^2} T_R \psi_{r+R}, \qquad T_{-R} = T_R^\dagger,$$

where only finitely many hopping matrices T_R are nonzero. The adjoint condition makes the operator Hermitian. Recall the positive-phase Bloch convention from *Quantum Mechanics I*, Lecture 8. Substitution of $\psi_r = e^{\mathrm{i}\mathbf{k}\cdot r} v$ gives a matrix acting on the orbital column.

Definition 25.1 (Bloch Hamiltonian)**.** The *Bloch Hamiltonian* in this convention is

$$H(\mathbf{k}) = \sum_R T_R e^{\mathrm{i}\mathbf{k}\cdot R}.$$

It is a smooth Hermitian $N \times N$ matrix. Adding 2π to either momentum component leaves this matrix unchanged.

Definition 25.2 (Brillouin torus)**.** The *Brillouin torus* is the momentum parameter space

$$\mathbb{T}^2 = \mathbb{R}^2/(2\pi\mathbb{Z})^2,$$

with orientation $\mathrm{d}k_x \wedge \mathrm{d}k_y$.

On an $L \times L$ periodic lattice, the allowed momenta have components $2\pi a/L$, with $a \in \{0, \dots, L-1\}$. The finite geometric sum makes Bloch waves with distinct momenta orthogonal. There are L^2 such momenta, each paired with the N orbital columns. The resulting Bloch waves form a basis of the NL^2-dimensional one-particle space. The substitution above therefore gives its full block decomposition. A finite sample therefore samples L^2 points of one smooth family $H(\mathbf{k})$. The family does not depend on L. Only the sampling changes. The eigenline bundle and its curvature belong to this continuous family. The infinite-lattice Bloch ansatz is not a normalized square-summable state.

Definition 25.3 (Band)**.** A *band* is an energy branch $E_n(\mathbf{k})$ of this family.

Definition 25.4 (Isolated band)**.** We call a chosen smooth simple band *isolated* when, at every momentum, its distance from every other eigenvalue is at least a fixed $\delta > 0$.

This is a separation at the same momentum. It does not yet assert an energy interval separating occupied and empty states at all momenta.

Physical input. We model a slowly driven electron by a wavepacket, a superposition of Bloch states concentrated near a central momentum $\mathbf{k}$ in an isolated band. For electron charge $-e$, with $e > 0$, a weak uniform electric field drives that momentum by

$$\hbar \dot{\mathbf{k}} = -e\mathbf{E}.$$

We assume interband transitions are negligible in this semiclassical approximation.

The field therefore changes the parameter of $H(\mathbf{k})$. Lecture 24's eigenline curvature can affect this slow evolution.

Proposition 25.5 (The Bloch bundle · *proved*)**.** *The eigenspaces of a smooth isolated simple band form a Hermitian line bundle $L_n \to \mathbb{T}^2$. For local unit eigenvectors $v_i(\mathbf{k})$, its connection is $\mathcal{A}_i = v_i^\dagger \mathrm{d}v_i$.*

Proof. The periodic matrix family is defined on the torus itself. Lecture 24's adjugate construction supplies smooth local eigenvectors there. Their unit-phase transitions give the line bundle and the projected Hermitian connection. Its curvature is global and imaginary. ∎

Definition 25.6 (Band Chern number)**.** The *band Chern number* of an isolated simple band is

$$C_n = \frac{\mathrm{i}}{2\pi} \int_{\mathbb{T}^2} \mathrm{d}\mathcal{A}.$$

The integral is real because the curvature is imaginary.

25.2 Integral flux and the two-band degree

Definition 25.7 (Smooth family)**.** A *smooth family* of Hermitian line bundles with connection here means one Hermitian line bundle with a smooth unitary connection on $\mathbb{T}^2 \times [0, 1]$, restricted to the momentum slices.

Proposition 25.8 (Integrality on the torus · *proved*)**.** *For a Hermitian line bundle with a unitary connection on the oriented torus, the number $\mathrm{i} \int_{\mathbb{T}^2} F/(2\pi)$ is an integer. It is unchanged in a smooth family as defined above.*

Proof. We cut the torus into a square and compare a unit frame across its paired edges. Stokes converts the curvature flux to edge-phase changes, as in Lecture 22. Compatibility at the corner forces their difference to be an integer multiple of 2π.

Pull the bundle and connection back along $\mathbb{R}^2 \to \mathbb{T}^2$. Transport a fixed unit vector at the origin along radial segments. Lecture 21's smooth parameter dependence gives a smooth unit frame on $\mathbb{R}^2$. Restrict it to the square $[0, 2\pi]^2$ and call its connection form A. Write $x = k_x$ and $y = k_y$ in this argument. After identifying equal torus fibres, the edge frames satisfy

$$e(2\pi, y) = e(0, y)h_x(y), \qquad e(x, 2\pi) = e(x, 0)h_y(x),$$

with smooth maps $h_x, h_y \colon [0, 2\pi] \to \mathrm{U}(1)$. Lecture 3's phase lifting on an interval gives real smooth functions χ_x and χ_y with $h_x = e^{\mathrm{i}\chi_x}$ and $h_y = e^{\mathrm{i}\chi_y}$. Smoothness follows locally by a smooth inverse of the exponential covering. Lecture 21 gives the frame-change law $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$. For $h = e^{\mathrm{i}\chi}$ the added term is $\mathrm{i}\mathrm{d}\chi$. Thus $A_R - A_L = \mathrm{i}\mathrm{d}\chi_x$ along the vertical edges and $A_T - A_B = \mathrm{i}\mathrm{d}\chi_y$ along the horizontal edges. We write e_L, e_R, e_B and e_T for the restrictions of the square frame to its four sides. The figure shows these edge comparisons after equal torus fibres have been identified.

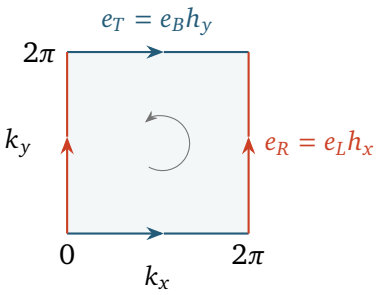


Figure 25.1: Opposite sides represent the same momenta. The phase functions compare the square frame across paired edges, and the circular arrow gives the positive boundary orientation.

Let $A_{R,y}$ and $A_{L,y}$ be the tangential coefficients on the vertical edges. Let $A_{T,x}$ and $A_{B,x}$ be those on the horizontal edges. The top and left sides have reversed boundary directions, so Stokes gives

$$\begin{aligned}\int_{\mathbb{T}^2} F = \oint_{\partial\square} A &= \int_0^{2\pi} (A_{R,y} - A_{L,y})\mathrm{d}y - \int_0^{2\pi} (A_{T,x} - A_{B,x})\mathrm{d}x \\ &= \mathrm{i}\Delta\chi_x - \mathrm{i}\Delta\chi_y,\end{aligned}$$

where $\Delta\chi = \chi(2\pi) - \chi(0)$. Consequently,

$$\int_{\mathbb{T}^2} F = \mathrm{i}(\Delta\chi_x - \Delta\chi_y), \qquad \frac{\mathrm{i}}{2\pi}\int_{\mathbb{T}^2} F = \frac{\Delta\chi_y - \Delta\chi_x}{2\pi}.$$

The right and top comparisons give the same frame at the last corner, so

$$h_y(0)h_x(2\pi) = h_x(0)h_y(2\pi).$$

Substitution of their lifted phases makes $\Delta\chi_x - \Delta\chi_y$ an integer multiple of 2π. This proves integrality. Neither edge phase is required to be a closed loop. In such a family, its global curvature restricts smoothly to the momentum slices, so the curvature integral varies continuously. An integer-valued continuous function on an interval is constant, proving the last claim. ∎

For a smooth Hamiltonian family whose chosen band stays isolated, the adjugate construction is smooth jointly in momentum and deformation parameter. The eigenline and projected connection of the chosen band give such a family.

For an example of integrality, the square potential $A = -\mathrm{i}nx\,\mathrm{d}y/(2\pi)$, with $n \in \mathbb{Z}$, has $h_x(y) = e^{-\mathrm{i}ny}$ and $h_y(x) = 1$. Its flux is $-2\pi\mathrm{i}n$, so its Chern number is n. The edge phase at $y = 2\pi$ agrees with its value at zero precisely because n is an integer. When $h_y = 1$, the corner identity makes h_x a closed loop. In that case $C = -\operatorname{wind}(h_x)$, as in Lecture 22's transition calculation.

A nonzero Chern number obstructs a smooth periodic unit eigenvector on the whole torus. Such an eigenvector would give a global form $\mathcal{A} = v^\dagger \mathrm{d}v$. Stokes on the closed torus would then give $\int_{\mathbb{T}^2} \mathrm{d}\mathcal{A} = 0$. Thus a nonzero number forces a phase mismatch between local eigenvectors.

Now take two orbitals per cell and a Hamiltonian $H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$, where $\mathbf{d}\colon \mathbb{T}^2 \to \mathbb{R}^3 \setminus \{0\}$ is smooth. The Pauli product identity gives energies $-|\mathbf{d}|$ and $+|\mathbf{d}|$.

Definition 25.9 (Direction map)**.** The *direction map* is given by

$$\widehat{\mathbf{d}}\colon \mathbb{T}^2 \longrightarrow S^2, \qquad \widehat{\mathbf{d}}(\mathbf{k}) = \frac{\mathbf{d}(\mathbf{k})}{|\mathbf{d}(\mathbf{k})|}.$$

Theorem 25.10 (The lower band as a degree · *proved*)**.** *For the lower band and the outward orientation on S^2,*

$$\mathrm{d}\mathcal{A} = -\frac{\mathrm{i}}{2}\widehat{\mathbf{d}} \cdot (\partial_{k_x}\widehat{\mathbf{d}} \times \partial_{k_y}\widehat{\mathbf{d}})\,\mathrm{d}k_x \wedge \mathrm{d}k_y, \qquad C = \deg(\widehat{\mathbf{d}}).$$

Proof. Dividing H by $|\mathbf{d}|$ changes its energies but leaves its eigenvectors unchanged. The lower eigenspace of $\widehat{\mathbf{d}} \cdot \boldsymbol{\sigma}$ is the excited eigenspace of Lecture 24's $-\mathbf{n} \cdot \boldsymbol{\sigma}$. That sphere line has curvature $-\mathrm{i}\beta/2$, where β is the outward area form. Indeed, its northern potential in Lecture 24 is $-\mathrm{i}(1 - \cos\theta)\mathrm{d}\phi/2$. Differentiating gives $-\mathrm{i}\sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi/2$. Lecture 12's area form satisfies $\beta_p(v, w) = p \cdot (v \times w)$ for $p \in S^2$ and $v, w \in T_pS^2$. Its pullback therefore has coefficient $\widehat{\mathbf{d}} \cdot (\partial_{k_x}\widehat{\mathbf{d}} \times \partial_{k_y}\widehat{\mathbf{d}})$. The band line is its pullback by $\widehat{\mathbf{d}}$. Pullback of the projected connection pulls back its curvature, giving the displayed formula. Lecture 14's degree identity gives

$$C = \frac{1}{4\pi}\int_{\mathbb{T}^2} \widehat{\mathbf{d}}^*\beta = \deg(\widehat{\mathbf{d}}),$$

because $\int_{S^2} \beta = 4\pi$. ∎

The degree counts preimages of a regular sphere direction with their orientation signs. It need not count a one-to-one covering of the sphere. The upper band has the opposite curvature and Chern number. For more than two orbitals, the curvature definition still applies, but this particular map to a direction sphere is unavailable.

25.3 The Qi–Wu–Zhang model

We seek a finite-range two-band lattice model with a direction map of nonzero degree. Finite-range hopping gives trigonometric polynomials. We choose $d_x = \sin k_x$ and $d_y = \sin k_y$. Their simultaneous zeros occur when both momentum coordinates belong to $\{0, \pi\}$. The third component can change sign at these four points as a parameter varies.

Definition 25.11 (Qi–Wu–Zhang model)**.** The *Qi–Wu–Zhang model*, abbreviated QWZ, has direction vector

$$\mathbf{d} = (\sin k_x, \sin k_y, u + \cos k_x + \cos k_y), \qquad u \in \mathbb{R}.$$

Its Hamiltonian is the matrix

$$H(\mathbf{k}) = \begin{pmatrix} u + \cos k_x + \cos k_y & \sin k_x - \mathrm{i} \sin k_y \\ \sin k_x + \mathrm{i} \sin k_y & -u - \cos k_x - \cos k_y \end{pmatrix}.$$

The parameter u is an energy parameter, distinct from Lecture 19's inverse radius. In the lattice convention of the first section its only possibly nonzero hopping matrices are $T_0 = u\sigma_z$, $T_{(1,0)} = (\sigma_z - \mathrm{i}\sigma_x)/2$ and $T_{(0,1)} = (\sigma_z - \mathrm{i}\sigma_y)/2$, together with their adjoints. Thus it is a finite-range Hermitian lattice system.

This is a model lattice Hamiltonian, rather than a model of a specified material. We call a momentum with $d_x = d_y = 0$ a *vertical point*. At such a point, the value of d_z is its *mass*. The mass multiplies σ_z and opens the local Dirac gap of §25.5. We denote the four points by Γ, X_1, X_2 and M, as in the table.

Example 25.12 (The QWZ phase diagram · *proved*)**.** Find the lower-band Chern number and every gap closing as u varies. The energies are $E_- = -|\mathbf{d}|$ and $E_+ = |\mathbf{d}|$, so a gap closes exactly when $\sin k_x = \sin k_y = 0$ and $u + \cos k_x + \cos k_y = 0$. The relevant masses and transverse Jacobian signs are

point	(k_x, k_y)	d_z	$\cos k_x \cos k_y$
Γ	$(0, 0)$	$u + 2$	$+1$
X_1	$(\pi, 0)$	u	-1
X_2	$(0, \pi)$	u	-1
M	(π, π)	$u - 2$	$+1$

Hence the critical parameters are -2, 0 and 2. There are two distinct gap-closing points at zero. At every other parameter, compactness of the torus gives a positive minimum gap.

To compute the degree, count preimages of the north pole. A row contributes precisely when its mass d_z is positive. Near such a point the first two components of $\widehat{\mathbf{d}}$ are local sphere coordinates. Their Jacobian determinant is $\cos k_x \cos k_y / d_z^2$ at the point. These coordinates have the outward sphere orientation at the north pole. All selected preimages are therefore regular, with the signs in the last column. Lecture 14's signed-count theorem gives

$$C = \begin{cases} 0, & u < -2, \\ +1, & -2 < u < 0, \\ -1, & 0 < u < 2, \\ 0, & u > 2. \end{cases}$$

For $u < -2$ there are no north preimages. In the next interval only Γ contributes. Crossing zero adds both negatively oriented points. Crossing two adds the positive contribution at M. Thus the simultaneous closing at zero changes the integer by -2.

The square and the embedded torus in the figure describe the same periodic coordinates.

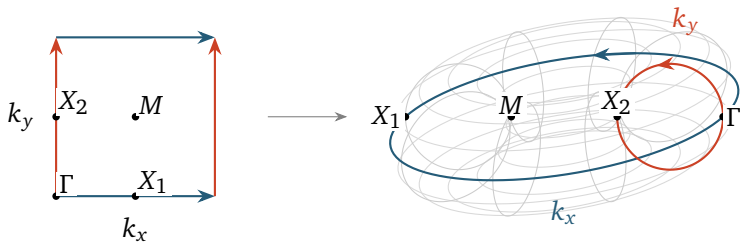


Figure 25.2: The parametrization $((2+\cos k_y)\cos k_x, (2+\cos k_y)\sin k_x, \sin k_y)$ embeds the momentum torus in $\mathbb{R}^3$. The marked points and the two coordinate circles agree with those on the square.

The vertical points and their orientation signs are now shown on the momentum square.

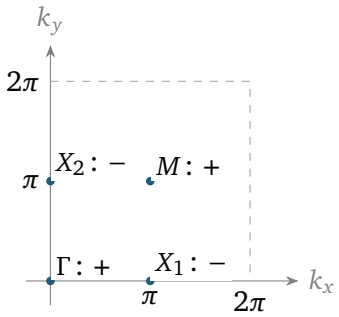


Figure 25.3: The four distinct vertical points in one torus cell. Each sign contributes to the north-pole count only when its mass is positive.

The direction maps show how these signed preimages give different degrees.

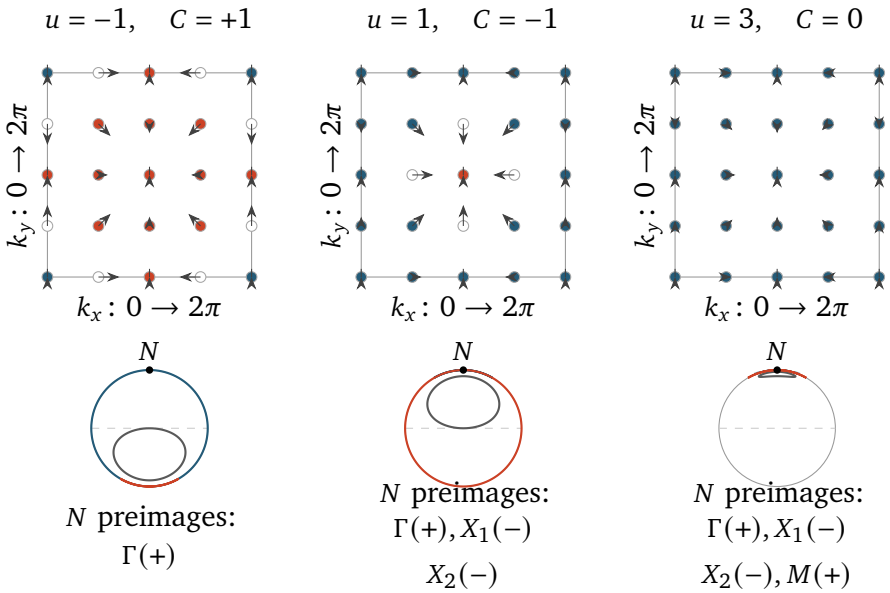


Figure 25.4: Arrows give $(\widehat{d}_x, \widehat{d}_y)$ on a grid with blue, red and white dots for positive, negative and zero $\widehat{d}_z$. The sphere curves show the images of $k_y = 0, \pi/2, \pi$ viewed along the y axis, with signed north-pole counts below each panel.

The phase diagram records the computed lower-band integer and its missing critical parameters.

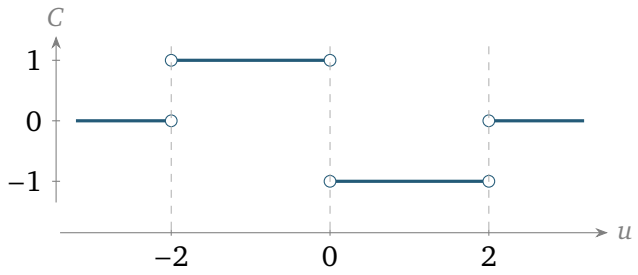


Figure 25.5: The QWZ lower-band Chern number as u increases. The number is undefined at $u = -2$, $u = 0$ and $u = 2$.

The band plots distinguish the two gapped examples from the critical parameters.

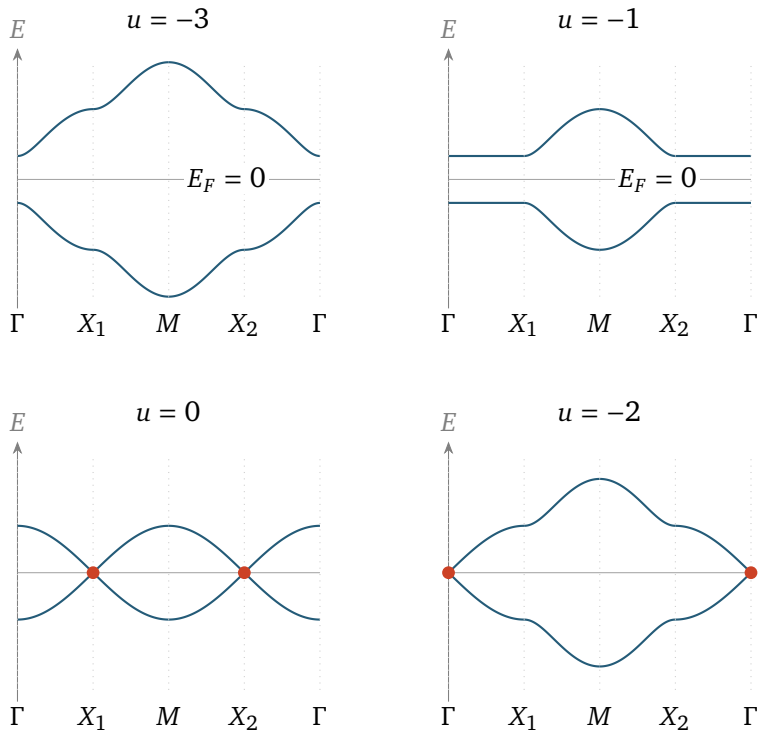


Figure 25.6: The energies E_- and E_+ along $\Gamma \to X_1 \to M \to X_2 \to \Gamma$ at four values of u. The closing at the repeated endpoint Γ is one torus point; the two X closings are distinct.

These values use $C = (\mathrm{i}/2\pi) \int \mathrm{d}\mathcal{A}$.[1]

The exact degree calculation requires no numerical mesh. An independent numerical check uses an $L \times L$ periodic momentum grid. Choose a unit lower-band vector w_j at each vertex, with arbitrary phase. Label a plaquette's four vertices counterclockwise. Multiply the forward overlaps $\langle w_1|w_2\rangle\, \langle w_2|w_3\rangle\, \langle w_3|w_4\rangle\, \langle w_4|w_1\rangle$ and divide each overlap by its magnitude, assuming none vanishes. Every vertex phase cancels from this product. With a

[1] Asbóth–Oroszlány–Pályi use the same real potential $A_\mathrm{B} = \mathrm{i}\mathcal{A}$. Their continuum definition is $Q = -(1/2\pi) \int \mathrm{d}A_\mathrm{B}$, so $Q = -C$ in their equations (2.38), (2.41) and (6.5).

counterclockwise plaquette, the product of normalized overlaps has argument approaching $\operatorname{Im} \int \mathrm{d}\mathcal{A}$. Thus the *negative* sum of these arguments, divided by 2π, approximates our C. At $u = -3, -1, 1, 3$, grids with 20×20 and 32×32 points give values close to $0, +1, -1, 0$, respectively, in agreement with a separate curvature integral. These numerical checks do not replace the degree proof.

Self-check. Compute the lower-band Chern number of the following Hamiltonian by counting south-pole preimages:

$$H_*(\mathbf{k}) = \mathbf{d}_*(\mathbf{k}) \cdot \boldsymbol{\sigma}, \qquad \mathbf{d}_* = (\sin(2k_x), \sin k_y, \frac{1}{2} + \cos(2k_x) + \cos k_y).$$

Determine the local signs using the outward orientation at the south pole.

Answer. The transverse components vanish for $k_x \in \{0, \pi/2, \pi, 3\pi/2\}$ and $k_y \in \{0, \pi\}$. The masses at these points are nonzero, so the band is isolated. Only $(\pi/2, \pi)$ and $(3\pi/2, \pi)$ have negative mass, equal to $-3/2$. At each of them the transverse Jacobian is $2\cos(2k_x)\cos k_y = 2$. The normalized chart Jacobian is therefore $2/(-3/2)^2 = 8/9$. The coordinates $(\widehat{d}_{*,x}, \widehat{d}_{*,y})$ have the opposite of the outward orientation at the south pole. Both local signs are -1, giving $C = -2$.

Self-check. Explain why the QWZ lower-band number is zero at $u = 3$, without counting preimages.

Answer. The component $d_z = 3 + \cos k_x + \cos k_y$ is at least one. Thus the direction map lies in the northern hemisphere and contracts there to the north pole. Its degree is zero by Lecture 14's homotopy invariance.

Unit 25B Hall response and gap closings

25.4 Filled bands and the Hall response

We consider a thin flat sample in the xy plane, with contacts that allow a transverse current. A voltage applied along x produces a weak field E_x. We measure the induced current per unit transverse length J_y. The *bulk* is the sample interior, away from its boundary. We call the periodic model *clean* when it has no disorder. Linear response is the term proportional to the applied weak field.

> **Definition 25.13** (Fermi energy)**.** The Fermi energy E_F is the zero-temperature occupation threshold.

Physical input. For noninteracting electrons at zero temperature, each one-particle state below E_F is occupied and each state above it is empty. Pauli exclusion permits one electron per state, with any spin already included among the orbitals.

> **Definition 25.14** (Band insulator)**.** An insulator has an open energy interval around E_F containing no bulk states.

This full energy gap is stronger than separation of a band at each momentum.

> **Definition 25.15** (Chern insulator)**.** A *Chern insulator* is a band insulator whose occupied bands have nonzero total Chern number.

For example, $H = 2\cos k_x I + \sigma_z$ has energies $2\cos k_x - 1$ and $2\cos k_x + 1$. Each band is isolated by a direct gap of two. Their energy ranges are $[-3, 1]$ and $[-1, 3]$, so no Fermi energy separates them completely. The Hamiltonian $H = \sigma_z$ instead has energies -1 and $+1$, with a full gap between them. The figure compares these periodic spectra, independent of k_y.

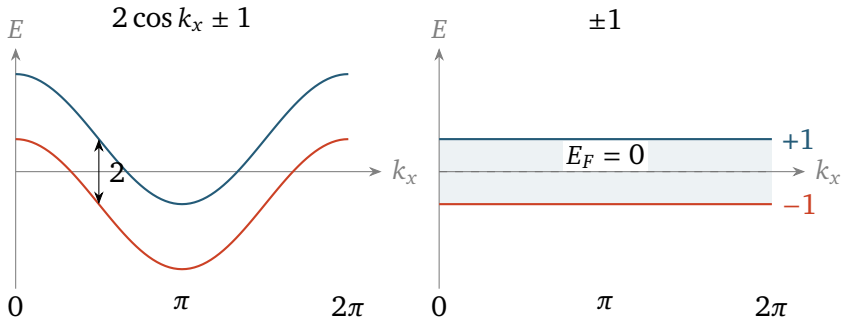


Figure 25.7: The spectra $2\cos k_x \pm 1$ overlap in energy despite their pointwise separation. The spectra ± 1 admit every Fermi energy in $(-1, 1)$.

The electron has signed charge $-e$, as fixed in §25.1. Write Planck's constant as $h_{\mathrm{P}} = 2\pi\hbar$, distinct from a frame-change map.

Definition 25.16 (Hall conductance). The scalar Hall conductance is defined by the transverse component convention

$$J_y = \sigma_{\mathrm{H}} E_x, \qquad J_x = -\sigma_{\mathrm{H}} E_y.$$

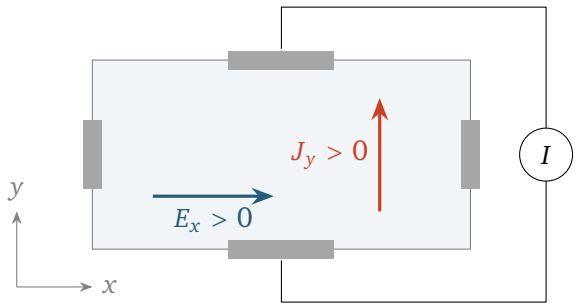
Physical input. **TKNN response 25.17** · *cited.* For a clean two-dimensional periodic noninteracting band insulator at zero temperature, with a bulk Fermi-energy gap and isolated occupied bands, the bulk linear Hall response is

$$\sigma_{\mathrm{H}} = \frac{e^2}{h_{\mathrm{P}}} \sum_{n \text{ occupied}} C_n.$$

For one filled band the sum has one term. In this ideal zero-temperature setting the longitudinal dc response vanishes.

We cite the filled-band response as TKNN. The formula also applies to periodic band insulators with zero magnetic flux through a unit cell, under the stated response hypotheses. The Kubo linear-response argument is cited rather than proved here (Tong, *The Quantum Hall Effect*, §2.3.1, equation (2.20), pp. 63–65). The response components follow the displayed definition. QWZ has no imposed uniform magnetic flux. The full bulk gap is compatible with the transverse filled-band response above. The integer stays fixed under deformations preserving the response hypotheses.

The sample sketch fixes the driving and measured directions for positive Hall conductance.



$$\sigma_H = Ce^2/h_{\mathrm{P}} > 0$$

Figure 25.8: A field along $+x$ produces transverse current along $+y$ when $C > 0$ for one occupied band. Transverse contacts measure the current indicated by the interior response arrows.

Physical input. For a boundary to a trivial insulator, bulk–boundary correspondence is additional cited physics (Asbóth–Oroszlány–Pályi, §6.2–6.3). For local noninteracting charge-conserving models, it fixes the net chirality of edge branches by the difference of occupied-band Chern numbers, with a fixed boundary orientation. Extra pairs of opposite-moving branches are possible. Edge-localized energies can cross E_F while the bulk gap remains open.

We derive the anomalous velocity as a non-rigorous first-order adiabatic approximation. We use a packet narrow in momentum, a smooth isolated nondegenerate band and slow driving. We omit higher-order terms and sustained transitions between bands.

Physical input. In the fixed orbital and cell-position convention of §25.1, the velocity operator is $\widehat{v}_a = \partial_a H/\hbar$, where $\partial_a = \partial/\partial k_a$ for $a \in \{x, y\}$. The filled-band current is the charge times the packet velocities summed with momentum density $\mathrm{d}^2k/(2\pi)^2$.

We choose local orthonormal eigenvectors $|n\rangle$ and $|m\rangle$. Along the path, we use a parallel gauge $\langle n|\dot{n}\rangle = 0$. After factoring the dynamical phase, the Berry phase is included by this gauge choice. We write the first-order state as

$$|\psi\rangle \simeq e^{-\mathrm{i}\int E_n \mathrm{d}t/\hbar}\left(|n\rangle + \sum_{m\neq n} c_m\,|m\rangle\right), \qquad c_m = \frac{\mathrm{i}\hbar\,\langle m|\dot{n}\rangle}{E_m - E_n}.$$

Indeed, projection of Schrödinger's equation onto $|m\rangle$ gives $(E_m - E_n)c_m = \mathrm{i}\hbar\,\langle m|\dot{n}\rangle$ to first order. Differentiating the eigenvalue equation and using orthogonality gives

$$\langle n|\,\partial_a H\,|m\rangle = (E_n - E_m)\,\langle \partial_a n|m\rangle\,.$$

Thus the energy denominators cancel in the velocity correction:

$$\begin{aligned}\hbar\langle\widehat{v}_a\rangle &\simeq \partial_a E_n + \sum_{m\neq n}\left(c_m\,\langle n|\,\partial_a H\,|m\rangle + \text{complex conjugate}\right)\\ &= \partial_a E_n - \mathrm{i}\hbar\left(\langle\partial_a n|\dot{n}\rangle - \langle\dot{n}|\partial_a n\rangle\right).\end{aligned}$$

Completeness supplies the last line. The omitted n term cancels in the antisymmetric difference because $\langle n|\partial_a n\rangle$ is imaginary. We define the real curvature components by

$$\Omega_{ab} = \mathrm{i}\left(\langle\partial_a n|\partial_b n\rangle - \langle\partial_b n|\partial_a n\rangle\right), \qquad a, b \in \{x, y\}.$$

The first-order velocity is therefore

$$\langle \widehat{v}_a \rangle \simeq \frac{\partial_a E_n}{\hbar} - \sum_{b \in \{x,y\}} \Omega_{ab} \dot{k}_b .$$

Let $\Omega_{\mathrm{B}} = \Omega_{xy}$, so $\mathrm{d}A_{\mathrm{B}} = \Omega_{\mathrm{B}} \mathrm{d}k_x \wedge \mathrm{d}k_y$. In two dimensions the second term is $-\dot{\mathbf{k}} \times (\Omega_{\mathrm{B}} \mathbf{e}_z)$, where $\mathbf{e}_z$ is the unit normal to the sample. For $E_x > 0$ and $\Omega_{\mathrm{B}} > 0$, the acceleration law gives negative $\dot{k}_x$. The anomalous electron velocity points along $-y$, so its negative charge produces current along $+y$. The figure separates momentum drift from real-space motion.

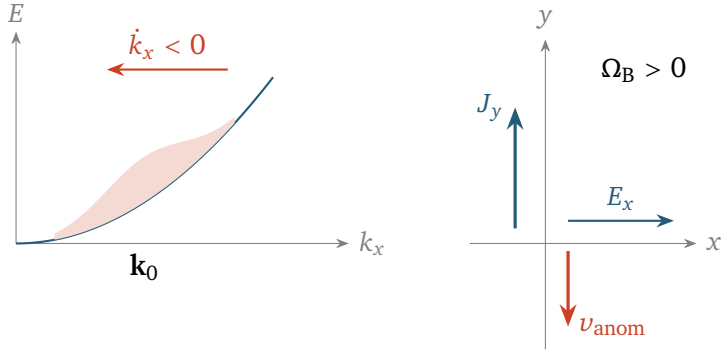


Figure 25.9: A packet near $\mathbf{k}_0$ drifts toward $-k_x$ under a positive E_x. For positive Ω_{B} its anomalous velocity is along $-y$, while its charge current is along $+y$.

Multiplication of this anomalous velocity by charge $-e$ gives a current contribution proportional to $(-E_y, E_x)$. For a filled band the group-velocity integral vanishes by periodicity:

$$\int_{\mathbb{T}^2} \partial_a E_n \mathrm{d}^2 k = 0, \qquad a \in \{x, y\}.$$

The momentum density of states is $\mathrm{d}^2k/(2\pi)^2$, so the anomalous contribution gives

$$\mathbf{J} = \frac{e^2}{\hbar} \int_{\mathbb{T}^2} \frac{\mathrm{d}^2 k}{(2\pi)^2} \Omega_{\mathrm{B}}(-E_y, E_x) = \frac{e^2}{h_{\mathrm{P}}} C(-E_y, E_x).$$

This checks the component sign and units of the cited formula. It is not a derivation of the Kubo theorem. For QWZ with only the lower band filled, the four insulating intervals have the conductances 0, $+e^2/h_{\mathrm{P}}$, $-e^2/h_{\mathrm{P}}$ and 0, in increasing u. Indeed, $\min|\mathbf{d}| > 0$ off the critical parameters, so $E_F = 0$ lies in the full gap between $-\min|\mathbf{d}|$ and $+\min|\mathbf{d}|$. At a critical parameter the gap hypothesis fails and the response law does not assign that quantized value.

25.5 Gap closings and Dirac cones

Definition 25.18 (Dirac cone). A *Dirac cone* is a linear two-band crossing with local Hamiltonian

$$H_{\mathrm{D}}(\mathbf{q}) = v_x q_x \sigma_x + v_y q_y \sigma_y + m\sigma_z, \qquad v_x v_y \neq 0.$$

Here $\mathbf{q}$ measures displacement from the crossing momentum. The energies of this local Hamiltonian at $m = 0$ are $\pm\sqrt{v_x^2 q_x^2 + v_y^2 q_y^2}$. The coefficients v_x and v_y are the local slopes, distinct from a unit eigenvector v. The mass m opens a gap. Integrating the lower-band

curvature density over a disk gives a finite-cutoff contribution. For $v_x = v_y = 1$ and $m \neq 0$, the contribution inside a circular disk is

$$C_{\text{cone}}(R) = \frac{1}{2}\operatorname{sgn}(m)\left(1 - \frac{|m|}{\sqrt{R^2 + m^2}}\right).$$

Here R is a momentum cutoff radius, rather than a lattice hopping vector. The figure shows this function against $R/|m|$.

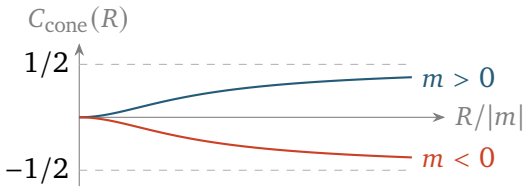


Figure 25.10: The continuum-cone contribution inside a disk of radius R, for unit positive slopes and $m \neq 0$. The limits are half-integers because the momentum plane is not a closed torus.

Proposition 25.19 (Changes at a closing · *sketched*)**.** *A smooth deformation of an isolated band cannot change its Chern number without losing its isolation. When isolated linear Dirac crossings are the only gap closings, their signed unit changes add. For a cone whose mass increases through zero, the change is* $\operatorname{sgn}(v_x v_y)$.

Sketch. Away from a closing, the eigenline curvature varies smoothly, so the integer remains constant by Proposition 25.8. Near a cone, the degree density gives the plane contribution

$$C_{\text{cone}} = \frac{1}{4\pi}\int_{\mathbb{R}^2} \frac{m v_x v_y \, \mathrm{d}q_x \mathrm{d}q_y}{(v_x^2 q_x^2 + v_y^2 q_y^2 + m^2)^{3/2}} = \frac{1}{2}\operatorname{sgn}(v_x v_y)\operatorname{sgn}(m).$$

Indeed, rescale the two momentum axes and use polar coordinates. The remaining radial integral is $\int_0^\infty |m| r (r^2 + m^2)^{-3/2} \mathrm{d}r = 1$. Its change across zero is the claimed signed unit. The curvature outside small crossing neighbourhoods has the same limiting integral from either side. In the local rescaling limit each neighbourhood has the displayed unit jump. Disjoint neighbourhoods add their contributions. This local limiting argument sketches the jump. It does not assign an integer to an isolated continuum cone. ∎

For QWZ the velocities at a vertical point are $v_x = \cos k_x$ and $v_y = \cos k_y$. Each local mass increases with u. The jumps are therefore +1 at −2, $-1 - 1 = -2$ at zero and +1 at two, in agreement with the full degree proof.

The direction map of $(v_x q_x, v_y q_y, m)$ maps the plane to an open hemisphere when $m \neq 0$. For positive $v_x v_y$, the north image at positive mass has signed area 2π. The south image at negative mass has signed area -2π. Division by 4π explains the two half-integers. The equator is approached only at infinite momentum. At $m = 0$, the origin is undefined and the punctured plane maps to the equator. The cone sections and their direction images are shown together.

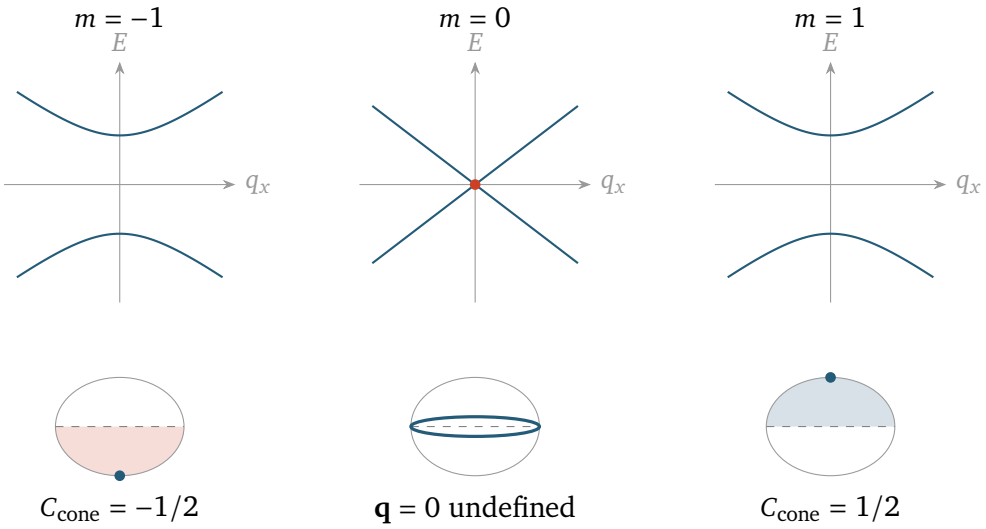


Figure 25.11: For unit positive slopes, the sections $q_y = 0$ have energies $\pm\sqrt{q_x^2 + m^2}$. The direction images are open hemispheres with signed flux $-1/2$ and $+1/2$ for negative and positive mass, and the equator with undefined origin at zero mass.

For QWZ, we can also view a closing in the parameter space $\mathbb{T}^2 \times [u_1, u_2]$. We give this space orientation $\mathrm{d}k_x \wedge \mathrm{d}k_y \wedge \mathrm{d}u$ and take gapped endpoint slices. The normalized pullback area form $\widehat{\mathbf{d}}^* \beta/(4\pi)$ is closed away from the zeros of $\mathbf{d}$. We remove small disjoint balls around its isolated zeros. Stokes gives

$$C(u_2) - C(u_1) = \sum_j \deg\left(\widehat{\mathbf{d}}\big|_{\partial B_j}\right).$$

The top slice has the positive torus orientation and the bottom has its negative. Each small sphere in this display has its outward orientation. Its orientation as an inner boundary of the remaining region is the opposite. This gives the sign in the display. Near a QWZ zero, the linear part is $(v_x q_x, v_y q_y, u - u_c)$. Its determinant is $v_x v_y$, so its small-sphere degree is $\operatorname{sgn}(v_x v_y)$. Higher-order terms do not change the degree on a sufficiently small sphere, by homotopy invariance. This is Lecture 24's parameter-space monopole applied to the lower eigenline of $H = \mathbf{d} \cdot \boldsymbol{\sigma}$. It is not a real-space magnetic monopole. The figure locates the four charges between the gapped slices.

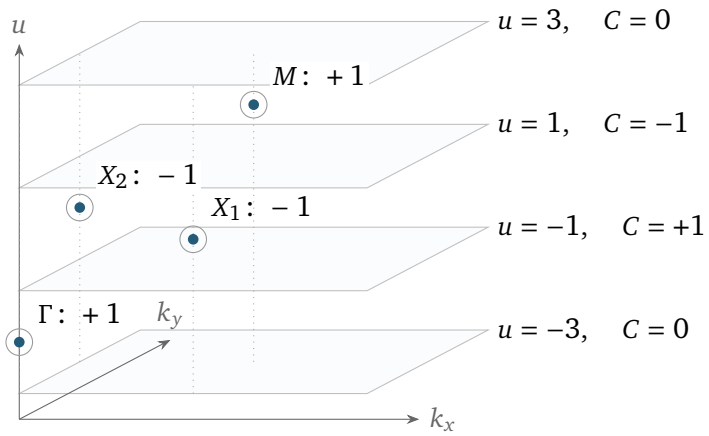


Figure 25.12: The axes have positive order (k_x, k_y, u), and each small sphere has its outward orientation. The four charges $+1, -1, -1, +1$ sum to the slice changes.

A deformation between different slice Chern numbers must therefore pass through a gap closing.

Self-check. State in one sentence what the band Chern number is and why it was introduced. Explain why a nonzero value prevents a smooth periodic unit eigenvector.

Answer. The band Chern number is the integer normalized curvature flux of an isolated eigenline over the momentum torus, introduced to classify gapped bands and determine their filled-band Hall response. A global periodic eigenvector would make $\mathcal{A}$ global, so Stokes on the closed torus would force that flux to vanish.

Summary. An isolated simple Bloch band defines a line bundle on the momentum torus. Its normalized curvature integral is an integer, fixed during any smooth deformation preserving band isolation. For $H = \mathbf{d} \cdot \boldsymbol{\sigma}$ with $\mathbf{d} \neq 0$, the lower-band number is $C = \deg(\widehat{\mathbf{d}})$. For QWZ its values are $0, +1, -1, 0$ in increasing u, with undefined numbers at $-2, 0, 2$. The cited filled-band response law gives the Hall conductance in units of e^2/h_{P}. The same principle of a homotopy class obstructing deformation will classify defects in Lecture 26.

Reading. Asbóth–Oroszlány–Pályi, *A Short Course on Topological Insulators*, §2.1.3–2.1.4 for the discrete sign; §2.2.1, §2.2.4 and §6.1, pp. 80–82 for QWZ, with continuum $Q = -C$. Thouless et al., *Quantized Hall Conductance in a Two-Dimensional Periodic Potential*, pp. 405–408, especially equation (5) on p. 406.